

Date	30 September 2026	Team ID	SWTID-2026-266
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer	Maximum Marks	4 Marks

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Catalog Item Ordering	- Hardware Catalog Navigation: Users can browse and select "Standard Laptop" in Hardware category. - Hardware Specs & Options: View technical specifications (Lenovo Carbon x1, i5, 512GB SSD) and choose optional software. - Order Checkout: Click "Order Now" to generate Request (REQ0010001) and Requested Item (RITM0010001).
FR-2	Approval Lifecycle Management	- Automatic Approver Routing: Generate pending approval record assigned to designated manager. - Single-Click Authorization: Manager can approve or reject the request under Approvers tab. - Stage Progression: Update RITM fulfillment stage and advance downstream automation upon approval.
FR-3	Flow Designer Automation Engine	- Trigger on Service Catalog: Detect approved catalog item requests without custom script logic. - System User Execution: Execute flow securely under System User in Global application scope. - Data Pill Mapping: Dynamically bind the triggering Requested Item record to target task record.
FR-4	Catalog Task Auto-Generation & Assignment	- Auto-Create Task: Create Catalog Task (sc_task) record linked directly to parent RITM. - Assignment Routing: Automatically assign record to "Hardware" assignment group. - Mandatory Fields: Pre-populate Short Description and Description as "Laptop need to Configured" and status as "Approved".
FR-5	Process Engine Configuration	- Catalog Item Binding: Map "Standard laptop task" flow directly to Standard Laptop in Maintain Items. - Legacy Workflow Removal: Detach deprecated workflows and execution plans under Process Engine tab.
FR-6	Hardware Staging & Fulfillment Tracking	- Queue Visibility: Technicians view new auto-generated tasks immediately in Hardware queue. - Work Notes & Closure: Technicians document imaging progress and close task to complete laptop handoff.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Clean, self-explanatory Service Catalog portal interface for end users with clear pricing and delivery estimates. Low-code, visual flow builder interface for administrators without complex background scripts.
NFR-2	Security	Adherence to ServiceNow Role-Based Access Control (RBAC). Only authorized managers can approve requisitions. Flow executes securely under the System User context within the Global application scope.
NFR-3	Reliability	Guaranteed, automated task creation for every approved laptop requisition, eliminating manual handoff omissions. ACID transactional integrity maintained across sc_request, sc_req_item, and sc_task tables.
NFR-4	Performance	Sub-second flow triggering upon manager approval; catalog task auto-generated within 3 seconds of approval event. Portal catalog item loading time under 2 seconds under normal workloads.
NFR-5	Availability	24/7 availability with 99.9% uptime aligned with ServiceNow cloud infrastructure, ensuring continuous self-service catalog ordering across enterprise global locations.
NFR-6	Scalability	Robust architecture capable of handling volume spikes during organizational refresh cycles without performance degradation. Modular design extensible to standard peripherals and monitor requests.

